



## CLIENT DELIVERABLE

# SAOS Dashboard Technical Specification

Systack Documentation

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| ROLE                   | COLOR       | HEX                                                                                           |
|------------------------|-------------|-----------------------------------------------------------------------------------------------|
| Headers, CTAs          | Navy        |  #001a2d |
| Navy Light             | Navy Light  |  #002845 |
| Secondary accents      | Teal        |  #007da9 |
| Primary buttons, links | Cyan        |  #00a1db |
| Gradients              | Cyan Bright |  #00c5e0 |
| Backgrounds            | Gray 50     |  #f8fafc |
| Cards                  | Gray 100    |  #f1f5f9 |
| Borders                | Gray 200    |  #e2e8f0 |
| Muted text             | Gray 400    |  #94a3b8 |
| Body text              | Gray 600    |  #475569 |
| Headings               | Gray 800    |  #1e293b |
| Success                | Green       |  #22c55e |

| ROLE              | COLOR  | HEX                                                                                         |
|-------------------|--------|---------------------------------------------------------------------------------------------|
| Error             | Red    |  #ef4444 |
| Accent highlights | Purple |  #8b5cf6 |

# SAOS Customer Dashboard – Technical Specification

**Document ID:** SYS-DASHBOARD-SPEC-v1.0 **Prepared for:** ORACLE (System Design Agent) **Prepared by:** SOL **Date:** June 25, 2026 **Status:** COMPLETE — Authoritative

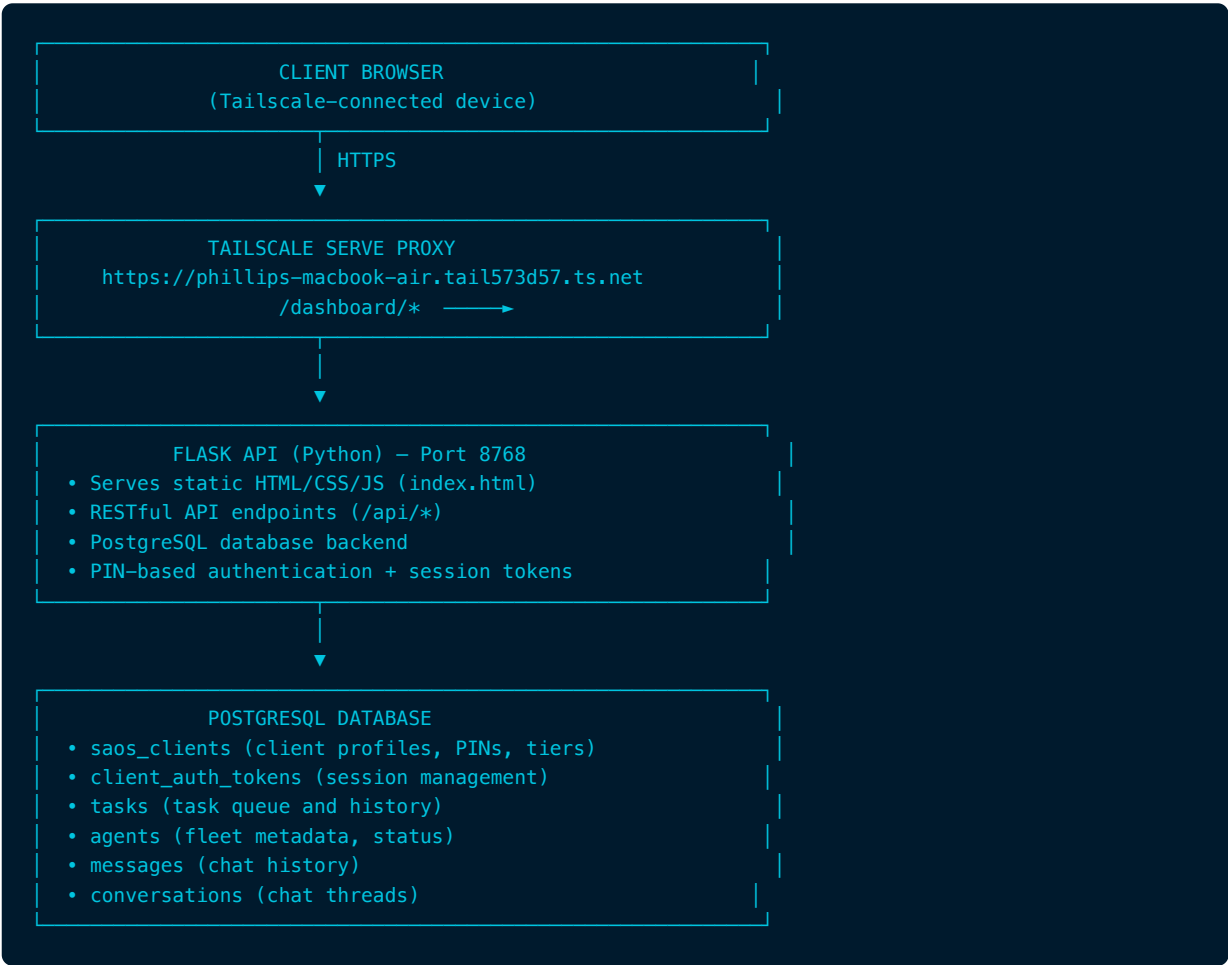
## Executive Summary

The SAOS Customer Dashboard is a **Flask-based web application** serving a single-page HTML frontend to SAOS Fleet subscribers. It provides:

- PIN-based authentication
- Real-time chat with AI agents
- Fleet status monitoring
- Task tracking
- Tier-specific service documentation
- Mobile-responsive design

**Access:** Tailscale VPN only (no public internet exposure)

## Architecture Overview



## File Structure

```
Systack/content/saos/saos-data/customer-dashboard/
├─ index.html           # Frontend SPA (119 KB, ~2,900 lines)
├─ api.py               # Flask backend (43 KB, ~1,100 lines)
├─ schema.sql           # Database schema (2.7 KB)
├─ admin_client.py      # CLI admin tool (6.3 KB)
├─ n8n-chat-bridge.json  # n8n chat integration workflow
├─ n8n-email-dispatcher.json # n8n email notification workflow
├─ deliverables/        # Client deliverables directory
│   └─ [task-specific files]
├─ README.md            # Basic overview
├─ migration_v2.1.sql    # DB migration script
├─ populate_task_descriptions.sql # Task seed data
├─ seed_agents.sql       # Agent seed data
├─
├─ # PDF Documentation (generated from markdown)
├─ SAOS-Service-Manual-v5.0.pdf # 393 KB (latest)
├─ SAOS-Dashboard-User-Guide-v3.0.pdf # 481 KB
├─ SAOS-Dashboard-Mobile-Access-Guide-v2.0.pdf # 281 KB
├─ SAOS-Quick-Start-Guide-v5.0.pdf # 279 KB
├─ SAOS-Architecture-Overview-v4.0.pdf # 429 KB
├─
├─ # Markdown Sources
├─ SAOS-Service-Manual-v5.0.md
├─ SAOS-Dashboard-User-Guide-v3.0.md
├─ SAOS-Dashboard-Mobile-Access-Guide-v2.0.md
├─ SAOS-Quick-Start-Guide-v5.0.md
├─ SAOS-Architecture-Overview-v4.0.md
├─
├─ # Backup files
├─ index.html.backup.20260624_052514
├─ index.html.broken
├─ index.html.fixed
```

## Frontend ( `index.html` )

### Technology Stack

- **Pure HTML/CSS/JavaScript** — No frameworks (no React, Vue, etc.)
- **CSS:** Custom properties (CSS variables) for theming
- **JS:** Vanilla JavaScript with fetch API for HTTP requests
- **No build step** — Direct file serving

### Design System

| TOKEN                        | VALUE                               | USAGE                     |
|------------------------------|-------------------------------------|---------------------------|
| <code>--bg-deep</code>       | <code>#001a2d</code>                | Page background           |
| <code>--bg-card</code>       | <code>rgba(255,255,255,0.04)</code> | Card backgrounds          |
| <code>--accent</code>        | <code>#00a1db</code>                | Primary buttons, links    |
| <code>--accent-bright</code> | <code>#00c5e0</code>                | Active states, highlights |
| <code>--text</code>          | <code>#e2e8f0</code>                | Primary text              |

| TOKEN                     | VALUE                | USAGE                   |
|---------------------------|----------------------|-------------------------|
| <code>--text-muted</code> | <code>#94a3b8</code> | Secondary text          |
| <code>--green</code>      | <code>#22c55e</code> | Success, online status  |
| <code>--amber</code>      | <code>#f59e0b</code> | Warning, pending status |
| <code>--red</code>        | <code>#ef4444</code> | Error, failed status    |
| <code>--purple</code>     | <code>#8b5cf6</code> | Accent gradients        |
| <code>--radius</code>     | <code>14px</code>    | Card corners            |
| <code>--radius-sm</code>  | <code>10px</code>    | Button corners          |

### Layout

- **Mobile-first responsive design**
- **CSS Grid + Flexbox** for layouts
- **Safe area insets** for iPhone notch ( `env(safe-area-inset-top)` )
- **Touch targets:** Minimum 44px for mobile buttons

### Application Structure

```
// Main app object (simplified)
window.App = {
  clientId: null,
  token: null,
  currentTab: 'dashboard',
  currentConversationId: null,

  init() { /* Initialize app */ },
  checkAuth() { /* Check localStorage for token */ },
  login(clientId, pin) { /* POST /api/auth/login */ },
  logout() { /* Clear localStorage + POST /api/auth/logout */ },

  // Tab handlers
  showTab(tabName) { /* Switch visible tab */ },
  renderDashboard() { /* Fleet status + metrics */ },
  renderChat() { /* Conversation list + message area */ },
  renderServices() { /* Tier-specific service cards */ },
  renderTasks() { /* Task history table */ },
  renderDocs() { /* Documentation iframe/links */ },
  renderActivity() { /* Activity log (v2.1) */ },

  // Chat
  loadConversations() { /* GET /api/chat/conversations */ },
  loadMessages(convId) { /* GET /api/chat/conversations/{id}/messages */ },
  sendMessage(text) { /* POST /api/chat/conversations/{id}/messages */ },
  pollForUpdates() { /* GET /api/chat/poll every 5s */ },

  // API helper
  apiFetch(endpoint, options) { /* Wrapper with auth header */ }
};
```

### Dynamic API Base Detection (Critical for Tailscale)

```
// Runtime detection of /dashboard prefix for reverse proxy
const API_BASE = (() => {
  const path = window.location.pathname;
  if (path.startsWith('/dashboard/')) return '/dashboard';
  if (path.startsWith('/dashboard')) return '/dashboard';
  return '';
})();

// All API calls use: `${API_BASE}/api/...`
// This prevents "The string did not match the expected pattern" errors
```

## Backend ( api.py )

### Technology Stack

- **Python 3** with Flask
- **Flask-CORS** for cross-origin requests
- **psycopg2** for PostgreSQL
- **No ORM** — Raw SQL with RealDictCursor

### Database Configuration

```
DB_HOST = os.environ.get("PGHOST", "localhost")
DB_PORT = int(os.environ.get("PGPORT", "5432"))
DB_NAME = os.environ.get("PGDATABASE", "systack_memory")
DB_USER = os.environ.get("PGUSER", "philliplowe")
```

### Authentication System

#### PIN-Based Login Flow

1. Client enters Client ID + PIN
2. POST /api/auth/login
3. Server verifies PIN hash in saos\_clients table
4. Generates secure token (secrets.token\_urlsafe(32))
5. Stores token hash in client\_auth\_tokens table
6. Returns token to client
7. Client stores in localStorage
8. All subsequent requests include: Authorization: Bearer <token>

### Session Management

- **Token expiry:** 30 days
- **Auto-renew:** On activity (last\_used\_at updated)
- **Storage:** Browser localStorage
- **Logout:** Clears localStorage + revokes token in DB

### API Endpoints

#### Authentication

| ENDPOINT           | METHOD | AUTH | DESCRIPTION                    |
|--------------------|--------|------|--------------------------------|
| /api/auth/register | POST   | No   | First-time client registration |
| /api/auth/login    | POST   | No   | Get Bearer token               |
| /api/auth/logout   | POST   | Yes  | Revoke token                   |
| /api/auth/me       | GET    | Yes  | Current client info            |

Portal (Fleet Data)

| ENDPOINT           | METHOD | AUTH | DESCRIPTION                    |
|--------------------|--------|------|--------------------------------|
| /api/portal/health | GET    | No   | Health check                   |
| /api/portal/status | GET    | Yes  | Fleet overview + client record |
| /api/portal/tasks  | GET    | Yes  | Task history (last 50)         |
| /api/portal/agents | GET    | No   | All agent states               |
| /api/portal/client | GET    | Yes  | Single client details          |

Chat

| ENDPOINT                              | METHOD | AUTH | DESCRIPTION         |
|---------------------------------------|--------|------|---------------------|
| /api/chat/conversations               | GET    | Yes  | List conversations  |
| /api/chat/conversations               | POST   | Yes  | Create conversation |
| /api/chat/conversations/{id}/messages | GET    | Yes  | Get messages        |
| /api/chat/conversations/{id}/messages | POST   | Yes  | Send message        |
| /api/chat/conversations/{id}/close    | POST   | Yes  | Close conversation  |
| /api/chat/poll                        | GET    | Yes  | Poll for updates    |
| /api/chat/webhook/agent-response      | POST   | No   | Webhook for agents  |

Admin (Protected)

| ENDPOINT           | METHOD | AUTH  | DESCRIPTION       |
|--------------------|--------|-------|-------------------|
| /api/admin/clients | GET    | Admin | List all clients  |
| /api/admin/clients | POST   | Admin | Create new client |

Database Schema

saos\_clients

```
CREATE TABLE saos_clients (  
  id SERIAL PRIMARY KEY,  
  client_id VARCHAR(50) UNIQUE NOT NULL, -- "TEST-001"  
  name VARCHAR(100) NOT NULL,  
  email VARCHAR(100),
```

```

pin_hash VARCHAR(64), -- SHA256 of PIN
tier VARCHAR(20), -- personal/business/enterprise/etc.
status VARCHAR(20) DEFAULT 'active',
created_at TIMESTAMP DEFAULT NOW(),
updated_at TIMESTAMP DEFAULT NOW()
);

```

### client\_auth\_tokens

```

CREATE TABLE client_auth_tokens (
  id SERIAL PRIMARY KEY,
  client_id INTEGER REFERENCES saos_clients(id),
  token_hash VARCHAR(64) NOT NULL, -- SHA256 of token
  created_at TIMESTAMP DEFAULT NOW(),
  expires_at TIMESTAMP DEFAULT NOW() + INTERVAL '30 days',
  last_used_at TIMESTAMP,
  revoked_at TIMESTAMP
);

```

### tasks

```

CREATE TABLE tasks (
  id SERIAL PRIMARY KEY,
  client_id INTEGER REFERENCES saos_clients(id),
  agent_name VARCHAR(50),
  task_type VARCHAR(50),
  description TEXT,
  status VARCHAR(20), -- pending/running/completed/failed/dead
  priority VARCHAR(20),
  created_at TIMESTAMP DEFAULT NOW(),
  completed_at TIMESTAMP
);

```

### messages + conversations

```

CREATE TABLE conversations (
  id SERIAL PRIMARY KEY,
  client_id INTEGER REFERENCES saos_clients(id),
  title VARCHAR(200),
  status VARCHAR(20) DEFAULT 'open',
  created_at TIMESTAMP DEFAULT NOW()
);

CREATE TABLE messages (
  id SERIAL PRIMARY KEY,
  conversation_id INTEGER REFERENCES conversations(id),
  sender_type VARCHAR(20), -- 'user' or 'agent'
  sender_name VARCHAR(50), -- agent name or 'You'
  content TEXT,
  created_at TIMESTAMP DEFAULT NOW()
);

```

---

## Network & Deployment

### Tailscale Configuration

```
# Tailscale serve config
tailscale serve https /dashboard/ http://localhost:8768
```

URL: `https://phillips-macbook-air.tail573d57.ts.net/dashboard/`

### LaunchAgent (Auto-Start)

File: `~/Library/LaunchAgents/net.systack.customer-dashboard.plist`

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN" "http://www.apple.com/DTDs/PropertyList-1.0.dtd">
<plist version="1.0">
<dict>
  <key>Label</key>
  <string>net.systack.customer-dashboard</string>
  <key>ProgramArguments</key>
  <array>
    <string>/usr/bin/python3</string>
    <string>/Users/philliplowe/.openclaw/workspaces/sol/Systack/content/saos/saos-data/customer-da
    <string>--port</string>
    <string>8768</string>
  </array>
  <key>RunAtLoad</key>
  <true/>
  <key>KeepAlive</key>
  <dict>
    <key>SuccessfulExit</key>
    <false/>
  </dict>
  <key>StandardErrorPath</key>
  <string>/tmp/customer-dashboard.err</string>
</dict>
</plist>
```

### Commands:

```
launchctl load ~/Library/LaunchAgents/net.systack.customer-dashboard.plist
launchctl unload ~/Library/LaunchAgents/net.systack.customer-dashboard.plist
launchctl list | grep customer-dashboard
```

## Service Tier Configuration

The dashboard adapts content based on client's `tier` field:

| TIER       | PRICE    | AGENTS | VPS     | SUPPORT   | DOCS ACCESS       |
|------------|----------|--------|---------|-----------|-------------------|
| personal   | \$49/mo  | 3      | Shared  | Email     | Base              |
| personal+  | \$99/mo  | 5      | Shared  | Priority  | Base              |
| business   | \$299/mo | 10     | 16GB    | Same-day  | Base + Enterprise |
| enterprise | \$799/mo | 10     | 32GB    | Dedicated | All               |
| private    | \$799/mo | 10     | On-prem | Dedicated | All + Compliance  |

| TIER       | PRICE    | AGENTS | VPS   | SUPPORT | DOCS ACCESS |
|------------|----------|--------|-------|---------|-------------|
| accelerate | \$249/mo | 5      | Cloud | Email   | Base        |

**Tier-specific content in Services tab:** - Business: Invoice processing, lead qualification, team Slack, reports - Enterprise: Above + on-premise option, HIPAA, white-glove, 4hr SLA - Private: Document extraction, compliance audit

## Known Issues & Gotchas

### 1. iOS Safari `.ts.net` Certificate Trust

**Issue:** iOS Safari doesn't trust Tailscale `.ts.net` certificates by default **Status:** OPEN — needs investigation **Workaround:** Use Tailscale app + “Use Tailscale DNS” setting

### 2. Reverse Proxy Path Prefix

**Issue:** When serving behind `/dashboard/` proxy, `fetch('/api/...')` resolves to root origin **Solution:** `API_BASE` dynamically detects prefix from `window.location.pathname` **File:** `index.html` line ~300

### 3. iOS Form Validation

**Issue:** iOS Safari blocks form submission with “The string did not match the expected pattern” **Solution:** Button changed to `type="button"` with `onClick` handler **File:** `index.html` login form

### 4. Mobile Layout

**Issue:** Navigation overflow on small screens **Solution:** Compact nav, larger sidebar toggle, shorter placeholders **File:** `index.html` CSS media queries

## Dependencies

### Python (for api.py)

```
flask>=2.0
flask-cors>=3.0
psycpg2-binary>=2.9
```

### External Services

- PostgreSQL (local or remote)
- Tailscale (for external access)
- n8n (optional, for chat bridge)

## Files Critical for ORACLE

When designing systems or researching solutions for the dashboard:

1. `index.html` — Frontend logic, API calls, auth flow, tab rendering

2. `api.py` — Backend endpoints, database queries, auth middleware
3. `schema.sql` — Database structure
4. `n8n-chat-bridge.json` — How chat messages route to agents
5. `admin_client.py` — How clients are provisioned

## Version History

| VERSION | DATE       | CHANGES                                                       |
|---------|------------|---------------------------------------------------------------|
| v1.0    | 2026-06-22 | Initial dashboard: Overview, Agents, Tasks, Account tabs      |
| v2.0    | 2026-06-24 | Complete rebuild: Chat, Dashboard, Services, Tasks, Docs tabs |
| v2.1    | 2026-06-25 | Added Activity tab, PIN auth, session tokens, mobile fixes    |

**Next Version:** v2.2 planned for iOS Safari cert fix + documentation updates

**Document Status:** AUTHORITATIVE — This is the single source of truth for dashboard technical details.

**Builder:** SOL 🦾 **Date:** June 25, 2026